

Title

Prediction and Visualization of the Human Hemoglobin Beta Protein Structure Using AlphaFold DB

Objective

The objective of this task was to predict and visualize the three-dimensional (3D) structure of the human hemoglobin beta protein using AlphaFold DB. The analysis aimed to examine the overall protein fold and identify major structural features.

Introduction

Protein structure is directly related to protein function. Predicting the three-dimensional structure of proteins helps researchers understand molecular interactions, stability, and biological activity. AlphaFold DB provides highly accurate computational models generated using artificial intelligence, making structural analysis accessible even when experimental structures are unavailable.

Materials and Methods

The amino acid sequence of human hemoglobin beta (UniProt accession **P68871**) was obtained from the UniProt database.

The predicted three-dimensional protein structure was retrieved from **AlphaFold Protein Structure Database**. The model was visualized using the interactive AlphaFold viewer, and the structure was examined from several different orientations.

Results

The predicted hemoglobin beta structure is dominated by **alpha-helices**, which is consistent with the known globin fold observed in vertebrate hemoglobin proteins.

The overall protein adopts a compact globular conformation composed of several long alpha-helices connected by short loop regions. Only a few flexible terminal regions are visible, indicating that most of the protein is highly ordered.

The AlphaFold model showed **high prediction confidence**, suggesting that the generated structure is reliable for structural interpretation.

Discussion

The predicted structure agrees well with the known architecture of globin proteins. The predominance of alpha-helical secondary structure is essential for maintaining the stability of the hemoglobin molecule and creating the binding pocket for the heme group.

The compact folding pattern reflects the highly conserved nature of hemoglobin beta among vertebrates. The visualization from different orientations provides a better understanding of the spatial arrangement of the helices and overall protein topology.

AlphaFold proved to be an efficient computational tool for predicting protein structures with high confidence.

Conclusion

The AlphaFold prediction successfully generated a reliable three-dimensional model of human hemoglobin beta.

The protein displays the characteristic globin fold dominated by alpha-helices and a compact tertiary structure. The predicted model is consistent with previously described structural features of hemoglobin proteins and demonstrates the usefulness of AI-based protein structure prediction.

Figure Captions

Figure 1. Three-dimensional structure of the human hemoglobin beta protein predicted using AlphaFold DB.

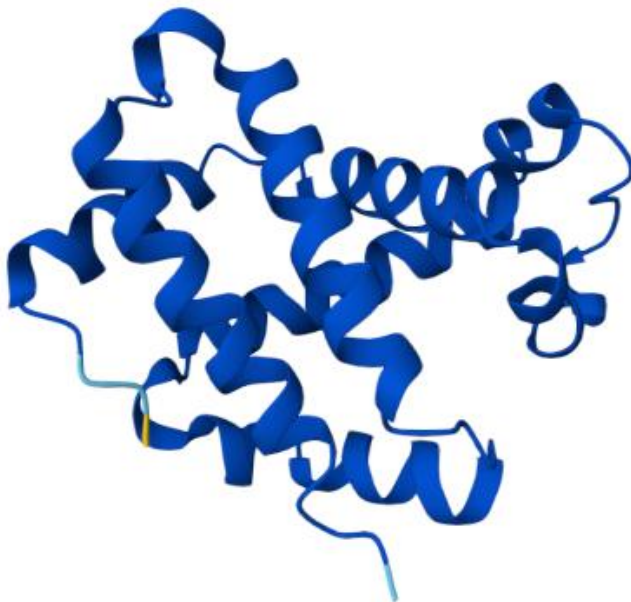


Figure 2. Rotated view of the predicted hemoglobin beta structure showing the spatial arrangement of alpha-helices.

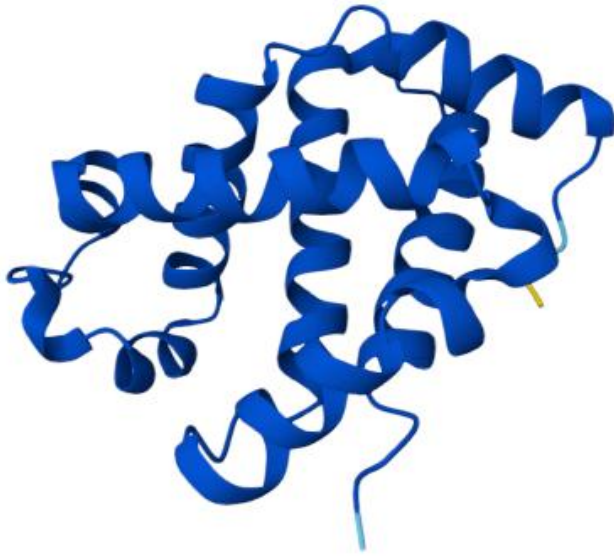


Figure 3. Alternative orientation of the protein highlighting its compact globular organization.

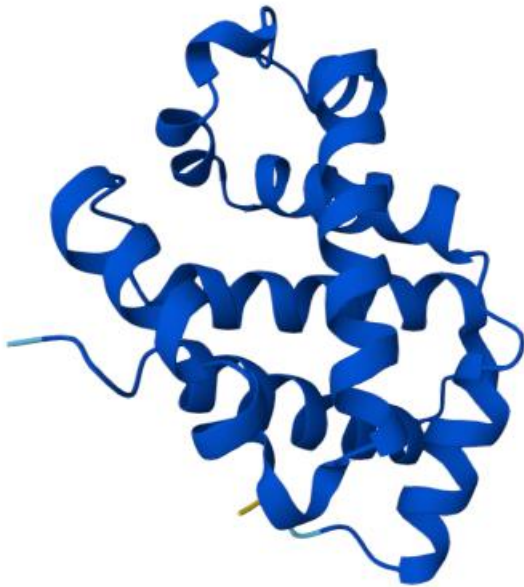


Figure 4. Additional structural view illustrating the overall folding pattern of the hemoglobin beta protein.